

Multivariable Linear Systems and Row Operations

Date_____ Period____

Write the augmented matrix for each system of linear equations.

1)
$$\begin{aligned} 5x - 2y &= -6 \\ -x + 5y &= 15 \end{aligned}$$

2)
$$\begin{aligned} -3x - 4y &= 20 \\ 3x - 5y &= 25 \end{aligned}$$

3)
$$\begin{aligned} x + 3y - 2z &= -11 \\ -2x - 5y + 3z &= 17 \\ 4x - z &= 1 \end{aligned}$$

4)
$$\begin{aligned} -2x - 4y - 5z &= 11 \\ -x + 4z &= -25 \\ -3x - 5y + z &= -25 \end{aligned}$$

Write the system of linear equations for each augmented matrix.

5)
$$\left[\begin{array}{cc|c} 3 & 4 & 1 \\ -3 & 2 & 23 \end{array} \right]$$

6)
$$\left[\begin{array}{cc|c} -5 & 1 & -16 \\ 1 & 5 & -2 \end{array} \right]$$

7)
$$\left[\begin{array}{ccc|c} 3 & -1 & 1 & 8 \\ 0 & -1 & 2 & -10 \\ -2 & 2 & 2 & -8 \end{array} \right]$$

8)
$$\left[\begin{array}{ccc|c} -5 & -4 & 3 & -8 \\ 1 & 0 & 4 & 0 \\ 3 & -5 & 5 & -10 \end{array} \right]$$